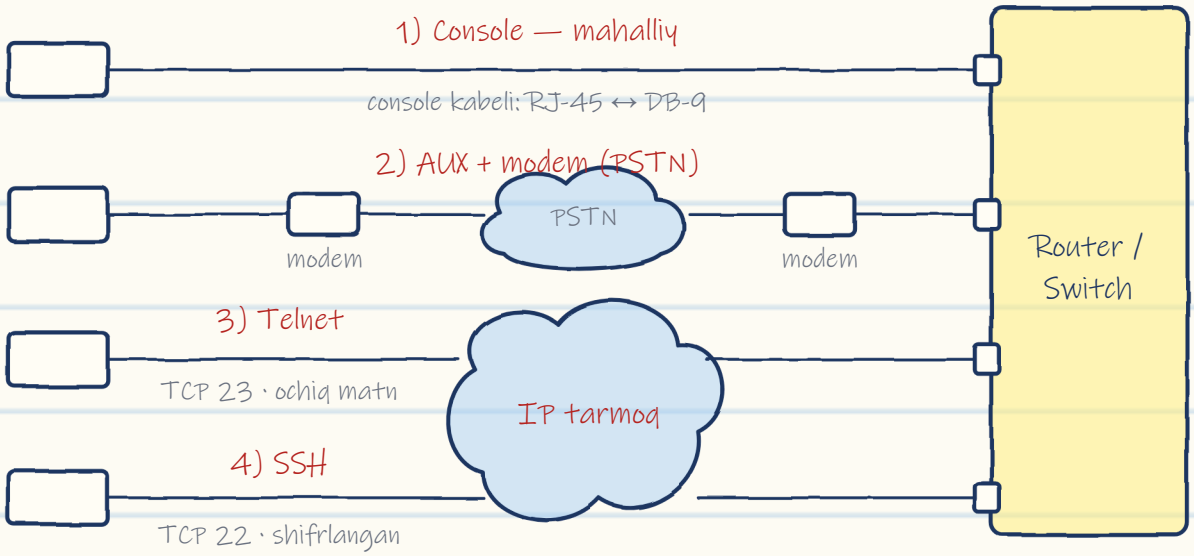
Buyruq satri bilan ishlash asoslari

Command Line Operation Basics

Asosiy atamalar
- CLI — buyruq satri, qurilmani boshqarish interfeysi
- Console port — kabel bilan mahalliy ulanish, 9600 bod
- Telnet / SSH — tarmoq orqali kirish: TCP 23 / TCP 22
- Buyruq ko'rinishi — user-view, system-view, interfeys-view

1. CLI ga ulanish usullari

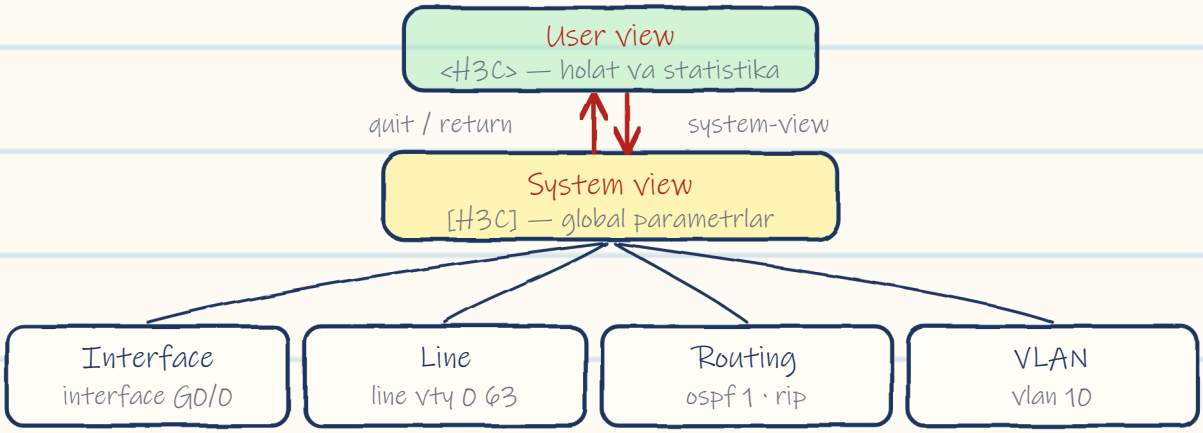
H3C Comware qurilmasi (CLI) — 4 ta asosiy kirish yo'li.



1-rasm. Console · AUX · Telnet · SSH — bitta qurilmaga to'rt yo'l

Usul	Port / protokol	Qayerdan
Console	EIA/TIA-232 DCE, 9600 bod	faqat mahalliy
AUX	EIA/TIA-232 DTE + modem	PSTN orqali
Telnet	TCP 23, ochiq matn	IP tarmoq
SSH	TCP 22, shifrlangan	IP tarmoq

2. Buyruq ko‘rinishlari (command views)



har qanday view’dan user view’ga: return yoki Ctrl+Z

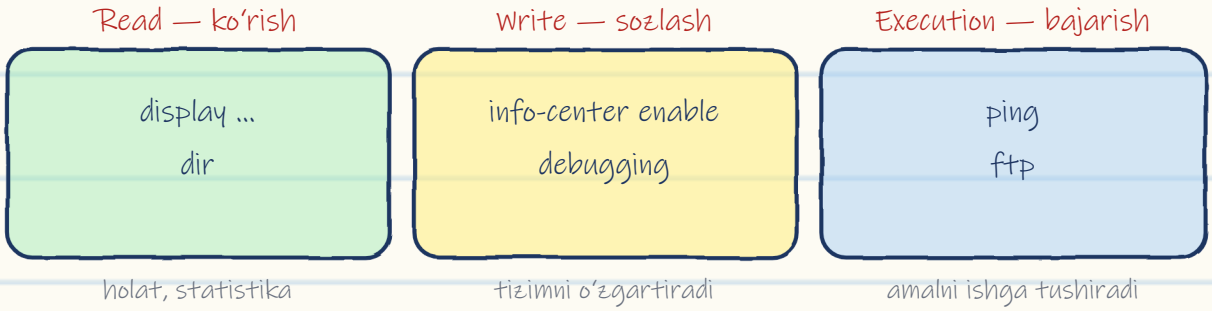
- Line view — console, AUX, TTY va VTY (telnet/SSH, 5 ta bir vaqtda)
- Pastga tushish — mos buyruq, yuqoriga — quit

3. Sessiya namunasi

```
<H3C> system-view
[H3C] interface GigabitEthernet 0/0
[H3C-GigabitEthernet0/0] description to_MyPC
[H3C-GigabitEthernet0/0] ip add 192.168.0.1 24
[H3C-GigabitEthernet0/0] quit
[H3C] line vty 0 63
[H3C-line-vty0-63] authentication-mode scheme
```

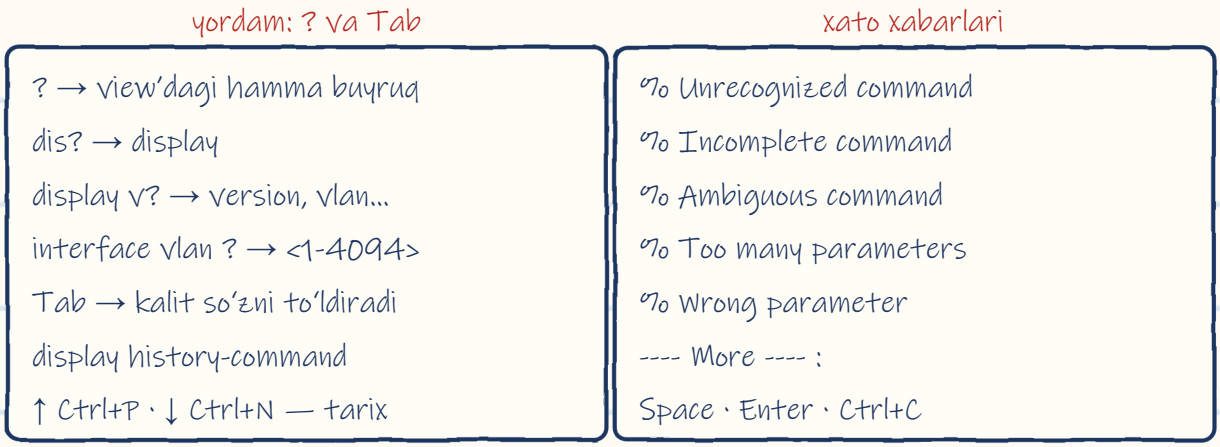
2-rasm. Prompt o‘zgarishi — qaysi view’dasiz, shu yozilgan

4. Buyruq turlari va rollar



network-admin	barcha funksiya va resurs
network-operator	faqat ko'rish (display history-command all — yo'q)
level-0 ... level-14	huquqni administrator beradi
level-15	network-admin bilan teng, o'zgartirilmaydi

5. Yordam, tarix, xatolar



3-rasm. Tarix buferi — 10 buyruq (history-command max-size)

6. Ko‘p ishlatiladigan buyruqlar

boshqarish

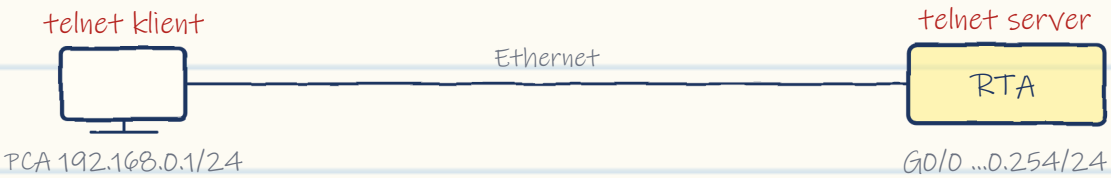
sysname RTA → <RTA>
clock datetime HH:MM:SS
display clock
header shell / login / motd
header legal / incoming
5 xil banner

ko‘rish

display version
display current-configuration
display interface
display ip interface brief
display diagnostic-information
hamma modul — bir buyruqda

Ctrl+A · Ctrl+E	satr boshi · oxiri	Ctrl+B · Ctrl+F	chapga · o‘ngga
Ctrl+D	kursordagi belgi	Backspace	chapdagi belgi
Ctrl+W · Esc+D	so‘zni o‘chirish	Ctrl+X · Ctrl+Y	chap · o‘ng tomon

7. Telnet serverni sozlash




```
[H3C] telnet server enable
[H3C] line vty 0 63
[H3C-line-vty0-63] authentication-mode password
[H3C-line-vty0-63] set authentication password simple 123456
[H3C-line-vty0-63] user-role network-admin
```

8. SSH serverni sozlash

parol bilan

login+parol shifrlanadi

serverni tekshirmaydi

ochiq kalit (RSA)

RSA kalit 512–2048 bit

SSH 2.0 → kamida 768 bit

```
public-key local create rsa
ssh server enable
line vty 0 63
authentication-mode scheme
protocol inbound ssh
local-user sshclient
service-type ssh
```

4-rasm. SSH: transport + autentifikatsiya + ulanish qatlamlari

Xulosa

- Kirish: console (mahalliy) · AUX · telnet 23 · SSH 22
- View'lar ierarxik: user → system → interface / line / ...
- return yoki Ctrl+Z — istalgan joydan user view'ga
- Buyruq turi: read · write · execution; rol: network-admin / operator / level-n
- ? va Tab — yordam, ↑↓ — tarix, Space/Enter/Ctrl+C — sahifalash

Tekshirish

1. SSH porti va telnet porti? — TCP 22 va TCP 23
2. Console tezligi (default)? — 9600 bod
3. Nechta VTY foydalanuvchi bir vaqtda? — 5
4. VTY autentifikatsiya rejimlari? — none / password / scheme
5. User view'ga qaytish? — return yoki Ctrl+Z